

LESSON 4.12

OPERATIONS WITH ABSOLUTE VALUE IN THE REAL WORLD



LESSON INTRODUCTION

MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute value.

LEARNING TARGETS

- I can perform operations that involve the absolute values of two numbers.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

M.A.7.NSO.2.1 Solve mathematical problems using multistep order of operations with rational numbers, including grouping symbols, whole-number exponents, and absolute value.

ESSENTIAL QUESTIONS

- How do you perform operations with absolute values in real-world contexts?
- How can number lines be used to create mathematical statements and perform operations with absolute values?

LESSON NARRATIVE

Students model various real-world situations with absolute value expressions. Students perform operations with absolute values to solve problems in context.

COMPONENT SUMMARY

4.12.1 WARM-UP (~5 MIN.)

Analyze a sale sign for an error

4.12.3 TRY IT (5 MIN.)

Practice writing and evaluating absolute value expressions in contexts

4.12.5 YOUR TURN! (10 MIN.)

Additional practice writing and evaluating absolute value expressions in contexts

4.12.2 GUIDED INSTRUCTION (10-15 MIN.)

Learn to write and evaluate absolute value expressions to solve problems

4.12.4 COLLABORATION (10 MIN.)

Collaboratively write and evaluate absolute value expressions in contexts

4.12.6 WRAP-UP (~5 MIN.)

Self-assessment of understanding of absolute value learning targets

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- (Optional) Chart paper & markers

ADDITIONAL PREPARATION

- Consider preparing images or videos of the contexts in the lesson to help students understand the contexts, as needed.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle with interpreting the written descriptions of real-world contexts to write absolute value expressions.
- Students may add unnecessary (and time-consuming) details to their number lines.

THINGS TO CONSIDER

- Students have not yet learned to perform operations with integers. Students should determine the absolute values first and then perform the operations.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

In 4.12.1 Warm-Up, students are asked what they notice and wonder about an image of a sale sign. Students should discuss what the error in the sale sign is and why it is not mathematically correct.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

In 4.12.5 Your Turn!, students interpret real-world situations to write absolute value expressions. Consider incorporating elements of students' personal experiences to enhance their learning and make meaning of absolute value in their own contexts.

DIFFERENTIATE INSTRUCTION

Consider providing sentence stems or answer choices for questions 4 and 5 in 4.12.4 Collaboration. This will help support students who struggle with expressing themselves in writing to demonstrate their understanding.

4.12.1 WARM-UP: MATH FAIL

Component Overview: Students demonstrate their mastery of the lesson learning targets. Use data from this formative assessment to determine which students need remediation before the next lesson.

TEACHER MOVES

Notice and Wonder

Ask students what they notice and wonder about an image of a sale sign. Students should discuss what the error in the sale sign is and why it is not mathematically correct.

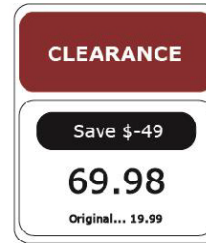
QUESTIONING

Consider challenging students to find more Math Fails as homework or extra credit as they review for the end of the unit.



4.12.1 Warm-Up: Math Fail

1. What do you notice about the sign?

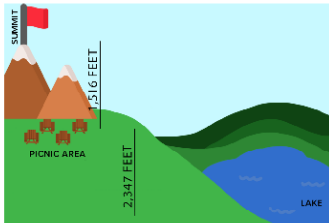


Answers will vary. The sale price is greater than the original price, so that's not a good sale. Also, the math is not correct because $69.98 - 49 = 20.98$.



4.12.2 Guided Instruction: Operations with Absolute Value in the Real World

1. Javier and his family drove to a picnic area on a mountain. From the picnic area, the family hiked to the mountain's summit, which was an elevation change of 1,516 feet (ft). They hiked back to the picnic area to eat lunch. After lunch, they hiked down to the lake, which was an elevation change of 2,347 ft.

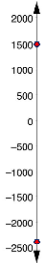


- a. In the table, write the value that represents the position of each location if the value of the position of the picnic area is 0. Then, plot each value at the estimated location on the number line.

Location	Value
Picnic area	0
Mountain summit	1516
Lake	-2347

- b. Represent the distance of each location from the picnic area.

Location	Distance from Picnic Area
Mountain summit	1516 feet
Lake	2347 feet



4.12.2 GUIDED INSTRUCTION: OPERATIONS WITH ABSOLUTE VALUE IN THE REAL WORLD

Component Overview: Students learn to write and evaluate absolute value expressions in contexts. The contexts explored include elevation and finance.

ENRICHMENT

- When calculating the distance between elevations, how do you know whether to add the absolute values or subtract them?
- How can the absolute values of a company's expenses and revenues be used to determine whether the company makes a profit?

4.12.2 GUIDED INSTRUCTION: OPERATIONS WITH ABSOLUTE VALUE IN THE REAL WORLD (CONTINUED)

QUESTIONING

Consider extending question 2 by asking early finishers the following:

Find the number of rooms Chloe needs to paint to break even.

How would the mathematical statement change if Chloe earns a fee of \$85 dollars per room?

- c. What operation would you use with the representations in the table to find the distance from the lake to the mountain summit? **Addition**
 - d. What is the distance from the lake to the mountain summit? **3863 feet**
 - e. What operation would you use to determine how much farther a hike to the lake is than a hike to the mountain summit? **Subtraction**
 - f. How much farther is a hike to the lake than a hike to the mountain summit? **831 feet**
2. Chloe is starting a painting business. The supplies she purchased for the business cost \$825. She will earn a fee of \$80 for each room that she paints. Chloe wants to determine how many rooms she will need to paint before she starts making a profit.
- a.

☐ Fees Chloe earns
☐ Supplies Chloe buys

 increase her profit.
 - b.

☐ Fees Chloe earns
☐ Supplies Chloe buys

 decrease her profit.
 - c. Write a value in the blank for each absolute value and a word in the remaining blank to complete each sentence.
 To find the revenue, Chloe multiplies the number of rooms she paints by **|80|**.
 To find the profit, Chloe subtracts **|-825|** from the **revenue**.
 - d. In the table, complete the mathematical statement and then perform the operations to determine if the number of rooms painted would result in a profit.

Number of Rooms	Mathematical Statement	Profit (Yes or No)
6	6 80 — -825	No
10	10 80 — 825	No
13	13 80 — -825	Yes
16	16 80 — -825	Yes



4.12.3 Try It: Operations with Absolute Value in the Real World

Logan, Mia, and Xavier all live on the same street as their school, which runs from east to west.

- Logan lives 4 blocks to the west of the school.
- Mia lives 5 blocks to the east of the school.
- Xavier lives 2 blocks to the west of the school.

1. Use the number line to model the position of each house and the school along the street. Label the location of each person's house and the school.



2. Work with your partner to write the mathematical sentence using absolute values. Then, use the mathematical sentence to determine each answer.
 - a. How far does Logan live from Mia?
 $| -4 | + | 5 | = 9 \text{ blocks}$
 - b. How far does Mia live from Xavier?
 $| 5 | + | -2 | = 7 \text{ blocks}$
 - c. How far does Xavier live from Logan?
 $| -4 | - | -2 | = 2 \text{ blocks}$
 - d. Was the same operation used to find each distance?
No

4.12.3 TRY IT: OPERATIONS WITH ABSOLUTE VALUE IN THE REAL WORLD

Component Overview: Students practice writing and evaluating absolute value expressions in contexts. The context explored is locations of students' homes relative to their school. Students use a number line to aid them in completing the questions.

ENRICHMENT

Sample questions to consider to check for understanding as students work:

- How did you determine where to plot the students' homes on the number line?
- How did you determine how far each student lives from school?
- How did you determine the distance between students' homes?
- How did you know whether to add or subtract the absolute values?

4.12.5 YOUR TURN!

Component Overview: Students practice labeling a number line with the elevations of the captain and his fishing net. Students create an absolute value expression to determine the length of the rope. Next, students write and evaluate absolute value expressions to represent debt and temperature changes of chicken in a freezer.

DIFFERENTIATE INSTRUCTION

Considerations to differentiate based on student need:

Facilitate small group conversations about what key points need to be plotted on the number line.

Review the number line from 4.12.2 Guided Instruction.

Prompt students to create an open number line with only the locations of 0, 23, and -38. Make sure students plot -38 farther from 0 than 23.

ENRICHMENT

- How is a fishing net used?
- How is a credit card used?
- What happens to the temperature of meat when you place it in the freezer?



4.12.4 Collaboration: Halfway Point

Mount Ojos del Salado is the highest mountain in Chile. Its peak is 6,893 meters above sea level. The lowest point of the Atacama Trench, just off the coast of Peru and Chile, is 8,065 meters below sea level.

Work with your partner to answer each question.

1. Determine how to label the number line so that you can approximate a location on the number line that represents Mount Ojos del Salado, Atacama Trench, and sea level.

Answers will vary. Answer shown on number line.

2. Label the number line and place a point at the approximate location for Mount Ojos del Salado, Atacama Trench, and sea level.

Answer shown on number line.

3. Write a mathematical sentence using absolute value to find the difference in elevations between Mount Ojos del Salado and the Atacama Trench.

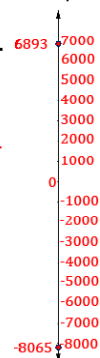
$$|-8065| + |6893| = 14,958$$

4. With your partner, discuss how the halfway point between the peak of Mount Ojos del Salado and the Atacama Trench can be found. Write down all of the strategies that were discussed.

Answers will vary. Divide the total distance by 2, and then add that number to -8,065 or subtract that number from 6,893.

5. Explain how to determine if the halfway point's elevation is above sea level or below sea level.

Answers will vary. The halfway point will be at an elevation below sea level because Atacama Trench is farther below sea level than Mount Ojos del Salado is above sea level.





4.12.5 Your Turn!

1. The captain of a fishing vessel is standing on the deck at 23 feet above sea level. He holds a rope tied to a fishing net below him that is underwater at a depth of 38 feet.
 - a. Draw a model of this situation using the number line.
Answer shown on number line.
 - b. Write a mathematical sentence using absolute value to determine the length of the rope.
 $|23| + |-38| = 61$
2. Aryan owes \$325.46 for recent purchases he made using his credit card, and he owes his parents another \$187.27.
 - a. Write a mathematical statement using absolute value that can be used to determine Aryan's total debt.
 $|-325.46| + |-187.27| = 512.73$
 - b. What is the value of Aryan's total debt?
 $\$512.73$
3. When Brenda places a package of chicken in the freezer, the chicken is at a temperature of 41°F . While in the freezer, the temperature of the chicken changes -2°F every hour until it reaches the temperature of the freezer. What is the temperature of the chicken 3 hours after Brenda puts it in the freezer?
 $|41| - 3|-2| = 35^{\circ}\text{F}$



4.12.6 WRAP-UP: REFLECTION

Component Overview: Students reflect on their mastery of the unit learning targets related to absolute value. Use the data from this formative assessment to determine which students need remediation before the next lesson.

DIFFERENTIATE INSTRUCTION

Consider which lessons to review based on students' responses:

- "I can compare the absolute value of two numbers." comes from Lesson 10.
- "I can perform operations that involve the absolute value of two numbers." comes from Lesson 10.
- "I can interpret comparisons of absolute values in real-world contexts." comes from Lesson 9.
- "I can perform operations that involve absolute value when solving real-world problems." comes from Lesson 12.